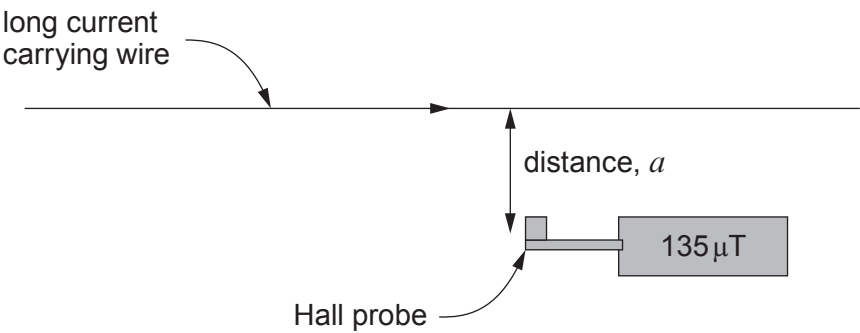


5. Michael carries out an experiment to determine how the magnetic field, B , due to a long current carrying wire varies with distance, a , from the wire.



His results are recorded in a table and a graph of $\ln(B)$ against $\ln(a)$ is plotted using the corrected B -field.

a/m	Measured B -field / T	Corrected B -field, B/T	$\ln(a/\text{m})$	$\ln(B/\text{T})$
0.010	0.000 529	0.000 499	-4.61	-7.60
0.020	0.000 279	0.000 249	-3.91	-8.30
0.040	0.000 157			
0.080	0.000 092	0.000 062	-2.53	-9.69
0.160	0.000 063	0.000 033	-1.83	-10.32
0.300	0.000 046	0.000 016	-1.20	-11.04

(a) Explain why the corrected B -field, B must be obtained. [1]

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(b) Complete the table above and plot the missing point on the graph opposite (no error bars are required). [3]

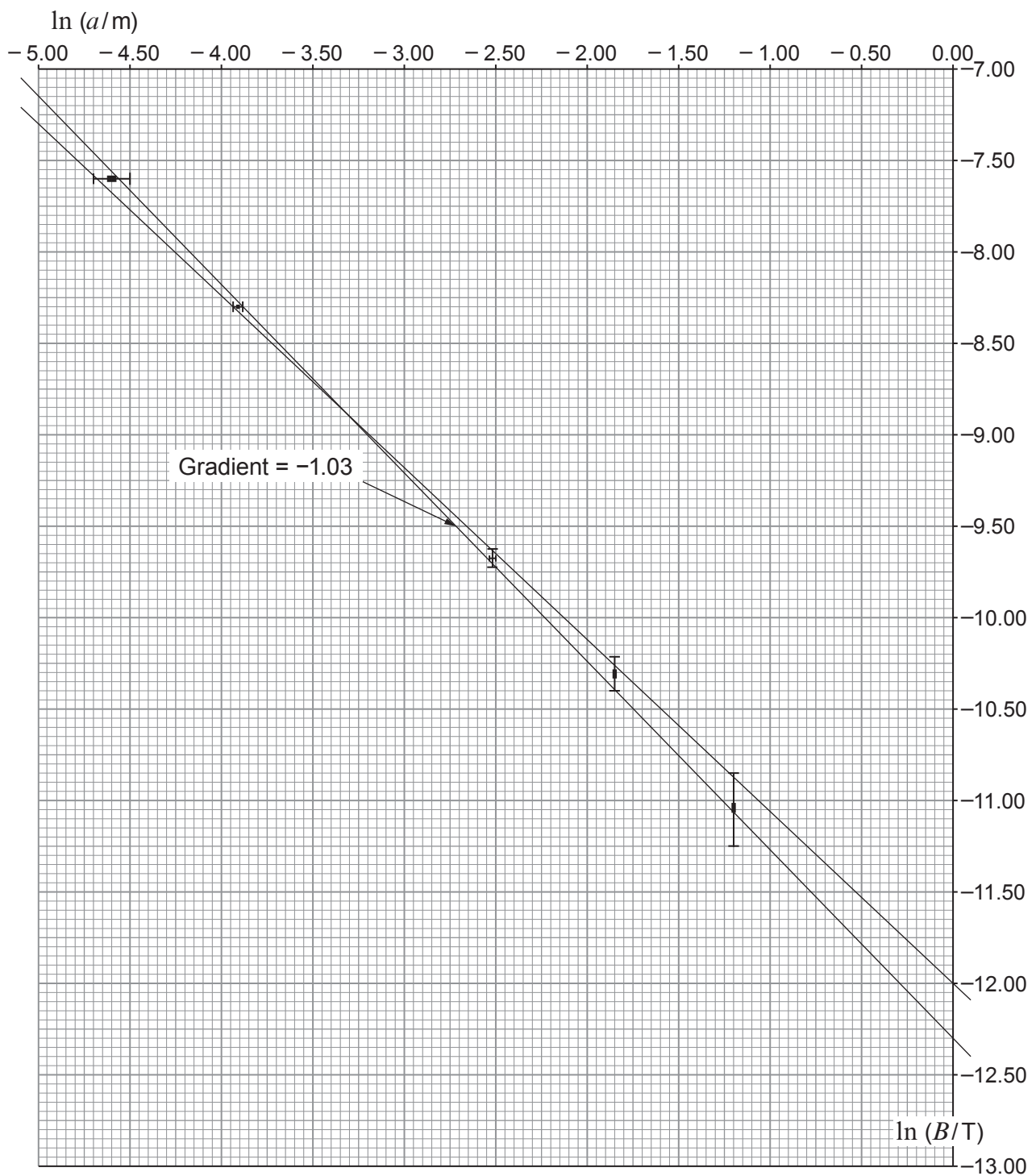
(c) The gradient of the steepest line is -1.03 . Calculate the gradient of the least steep line. [2]

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- (d) Use the equation $B = \frac{\mu_0 I}{2\pi a}$ to show that the expected mean gradient is -1 and that the expected $\ln(B)$ intercept is $\ln\left(\frac{\mu_0 I}{2\pi}\right)$. [2]

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- (e) Explain to what extent the graph confirms the relationship $B = \frac{\mu_0 I}{2\pi a}$. [3]

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- (f) Calculate the current in the wire along with its **absolute** uncertainty. [5]

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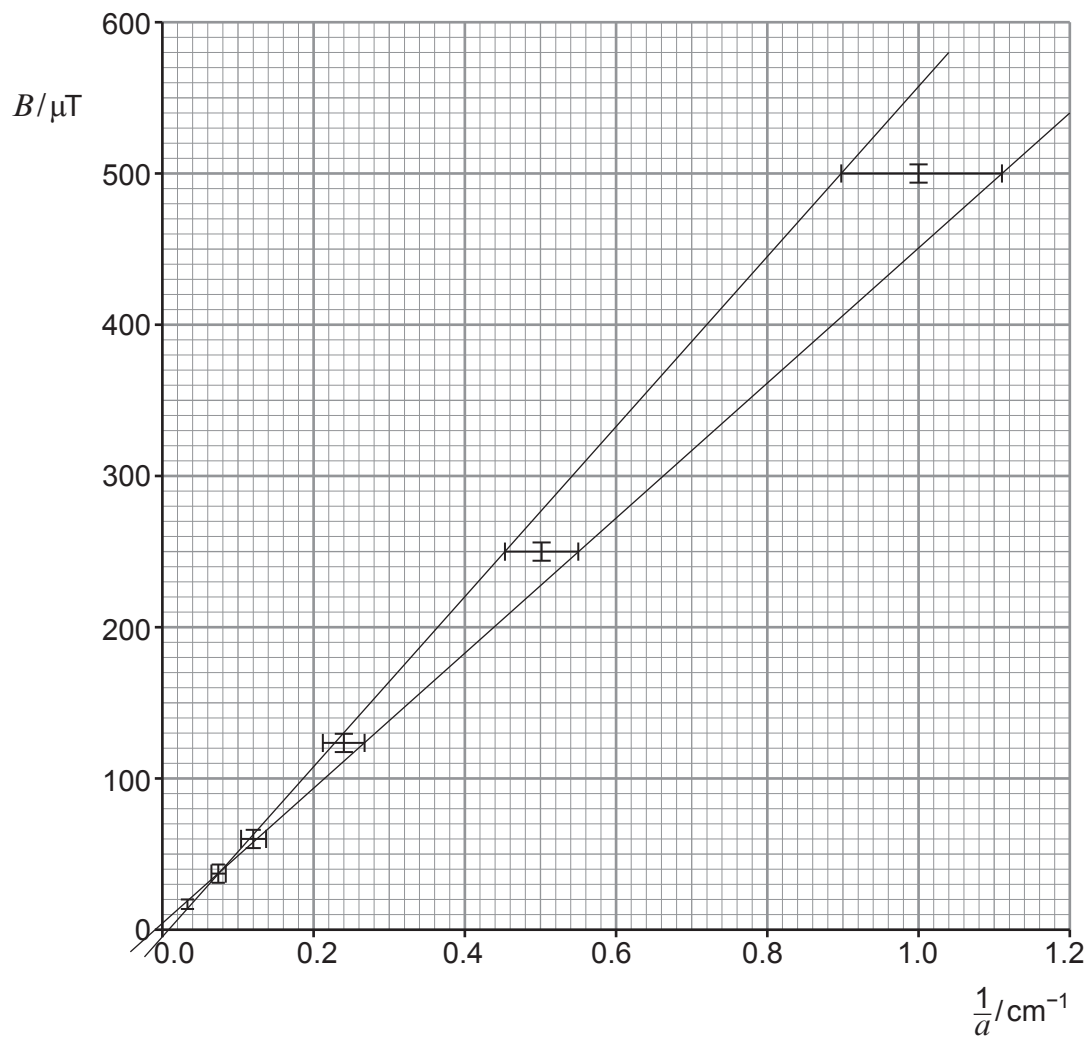
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(g) Bethan analyses Michael's data by plotting a graph of B against $\frac{1}{a}$.



Explain briefly why Michael's log graph is more suitable than Bethan's graph for this data set (no calculations are required).

[2]

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